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% EEL3135 Lab 3
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% Part 1 -
% Calculate the phasors using Matlab in polar form for each set of Fourier
coefficients.
%The code should use a for-loop where the amplitude and phase are held in
the vectors, Amp and Phi (provide
%the code and solution).
%Amp = [A-9, A-8, A-7, A-6, A-5, A-4, A-3, A-2, A-1, A0, A1, A2, A3, A4, A5,
A6, A7, A8, A9]
%Phi = [ϕ-9, ϕ-8, ϕ-7, ϕ-6, ϕ-5, ϕ-4, ϕ-3, ϕ-2, ϕ-1, ϕ0, ϕ1, ϕ2, ϕ3, ϕ4, ϕ5,
ϕ6, ϕ7, ϕ8, ϕ9]

data =  [3-2.5*i, 0+0*i, 2-2.5*i, 0+0*i, 0+0*i, 3-3*i, 0+0*i, 8-5*i, 0+0*i,
5, 0+0*i, 8+5*i, 0+0*i, 3+3*i, 2+2.5*i, 0+0*i, 2+2.5*i, 0+0*i, 3+2.5*i]

N = length(data);
Phasor = zeros(1, N);
Amp = zeros(1,N);
Phi = zeros(1,N);

for count = 1:N
    Amp(count) = abs(data(count));
    Phi(count) = angle(data(count));
    Phasor(count) = Amp(count) * exp(1j * Phi(count));
end

% Part 2 -
% Provide the MatLab code for the time vector, "t" that will be used in the
Fourier synthesis equation.
% This will be held as a vector and used to build your sinusoid components.
fs = 80

dt = 1/fs

t = 0:dt:1-dt

% Part 3 -
% Create another for loop that will loop through the ak fourier coefficients
which will create each sinusoidal
% component Xk(t).
% These sinusoidal components will be held in a Matrix, M(K, N), where k is
the number of % fourier coefficients and N is the length of the signal.
% for ?
%     M(?, ?) = ?
% end

% Part 4 % Provide the code for how the Fourier synthesis will be
implemented. % Part 5 -
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% Plot 3 rows of the Matrix that are tied to an ak. Describe the signals for
each plot and its relationship to the %components of a sinusoidal wave.
% plot ?
% Part 6 -
% Add the Matrix rows to then construct the X(t)Sum signal, using the matlab
function "sum". % X_t_sum

% Part 7 -
% What is the DC offset of the synthesized signal and how does this relate
to the fourier coefficients?

% Part 8 -
% Provide a plot of the synthesized signal. Use the grid on feature on the
plot and describe what the graph %represents.
% plot ?
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data =   Columns 1 through 4   3.0000 - 2.5000i   0.0000 + 0.0000i   2.0000 - 2.5000i   0.0000 + 0.0000i

Columns 5 through 8   0.0000 + 0.0000i   3.0000 - 3.0000i   0.0000 + 0.0000i   8.0000 - 5.0000i

Columns 9 through 12   0.0000 + 0.0000i   5.0000 + 0.0000i   0.0000 + 0.0000i   8.0000 + 5.0000i

Columns 13 through 16   0.0000 + 0.0000i   3.0000 + 3.0000i   2.0000 + 2.5000i   0.0000 + 0.0000i

Columns 17 through 19   2.0000 + 2.5000i   0.0000 + 0.0000i   3.0000 + 2.5000i

fs =

80

dt =

0.0125
t =   Columns 1 through 7   0   0.0125   0.0250   0.0375   0.0500   0.0625   0.0750

Columns 8 through 14   0.0875   0.1000   0.1125   0.1250

0.1375   0.1500   0.1625

Columns 15 through 21   0.1750   0.1875   0.2000   0.2125

0.2250   0.2375   0.2500

Columns 22 through 28   0.2625   0.2750   0.2875   0.3000

0.3125   0.3250   0.3375

Columns 29 through 35   0.3500   0.3625   0.3750   0.3875

0.4000   0.4125   0.4250

Columns 36 through 42   0.4375   0.4500   0.4625   0.4750

0.4875   0.5000   0.5125

Columns 43 through 49   0.5250   0.5375   0.5500   0.5625

0.5750   0.5875   0.6000

Columns 50 through 56   0.6125   0.6250   0.6375   0.6500

0.6625   0.6750   0.6875

Columns 57 through 63   0.7000   0.7125   0.7250   0.7375

0.7500   0.7625   0.7750

Columns 64 through 70   0.7875   0.8000   0.8125   0.8250

0.8375   0.8500   0.8625

Columns 71 through 77   0.8750   0.8875   0.9000   0.9125

0.9250   0.9375   0.9500

Columns 78 through 80

0.9625   0.9750   0.9875

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